

1 Target Input

Source Code

```
private void error() {  
    org.hibernate.Transaction txn =  
        session.getTransaction();  
    if (txn != null &&  
        txn.getStatus().equals(  
            TransactionStatus.ACTIVE)) {  
        txn.rollback();  
    }  
}
```

Baseline Summary (Generated by EP4CS)

"Logs an error message."

Why this baseline summary fails:

It hallucinates logging behavior and omits the guarded transaction rollback.

2 MOCSR Refinement Context

Structured Code Evidence

Coherence Evidence

Flow: getTransaction() -> active-state check -> rollback() on the true branch.

Consistency Evidence

Signature: error() -> void; guarded update: txn.rollback().

Relevance Evidence

Supporting facts: obtain the current transaction and roll it back only when active.

Retrieved Similar Examples (top-1 from each strategy)

Token-based example | Jaccard = 0.441

```
Transaction tx = manager.getTransaction();  
if (tx != null && tx.getStatus()  
    != Status.STATUS_NO_TRANSACTION) return tx;  
else return null;
```

Reference Summary: Checks whether or not we have an active transaction . If so returns it .

Semantic-based example | cosine = 0.780

```
Transaction tx = currentTransaction();  
if (tx == null)  
    throw new IllegalStateException("No active transaction");  
tx.rollback();
```

Reference Summary: Rolls back the current transaction if one exists .

3 Refinement Result

Model Decision

REVISE

Refined Summary

"Rolls back the current transaction if it is active."

Reference Summary (Evaluation Only)

"If transaction is present and active then rollback."

CodeRPE Scores (0--4)

Coherence 1 → 4

Consistency 0 → 4

Fluency 1 → 4

Relevance 1 → 4

Mean: 0.75 → 4.00

Automatic Metrics

BLEU 2.05 → 3.51

METEOR 5.81 → 27.47

ROUGE-L 0.00 → 35.29